

EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 2

PRINTABLE PRACTICE WORKSHEET

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Candidate Name:		Roll No.:	
Class / Grade:		Date:	
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Subject	Biology	Topic	Biology Comprehensive Practice Test 2	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. The first line of defence against many pathogens includes:

- A.** Skin and mucous barriers
C. Memory B cells only

- B.** Antibodies only
D. Red blood cells

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

2. Innate immunity is best described as:

- A.** A rapid, broadly acting defence present before a specific infection is encountered
C. A defence based entirely on antibodies

- B.** A response that occurs only after vaccination
D. A system that has no cells

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

3. Antibodies are produced by:

- A.** Activated B-lineage cells, especially plasma cells
C. Platelets

- B.** Red blood cells
D. Alveolar cells

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

4. A key feature of adaptive immunity is:

- A.** Specific recognition and immunological memory
C. Immediate action without recognition

- B.** Absence of receptors
D. Inability to respond again

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

5. Vaccination works primarily by:

- A.** Training the immune system to recognise specific targets without causing the intended disease
C. Replacing white blood cells

- B.** Destroying every microorganism in the body
D. Making pathogens unable to mutate

Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

6. A booster dose can be useful because:
A. Immune memory can be strengthened and antibody responses can rise again
C. It changes all pathogens into harmless organisms
Answer: ☐A ☐B ☐C ☐D

7. A person infected with a virus is not necessarily helped by an antibiotic because:
A. Antibiotics target bacterial structures/processes rather than viral replication mechanisms
C. Antibiotics only work on plants
Answer: ☐A ☐B ☐C ☐D

8. Antimicrobial resistance is strongly promoted when:
A. Susceptible microbes are killed while resistant variants survive and reproduce under selection pressure
C. Antibiotics create every mutation deliberately
Answer: ☐A ☐B ☐C ☐D

9. Autoimmune disease involves:
A. Immune responses directed against the body's own tissues
C. Only bacterial infection
Answer: ☐A ☐B ☐C ☐D

10. An allergy is best understood as:
A. An excessive or inappropriate immune response to a usually harmless substance
C. A lack of antibodies in every case
Answer: ☐A ☐B ☐C ☐D

11. Which immune cell type can directly kill infected or abnormal target cells?
A. Cytotoxic T cells
C. Platelets
Answer: ☐A ☐B ☐C ☐D

12. Inflammation can be useful because it:
A. Helps recruit immune cells and increases local defensive responses
C. Prevents all immune responses
Answer: ☐A ☐B ☐C ☐D

13. Why can a second exposure to the same antigen sometimes produce a faster response?
A. Memory cells formed during the first response can respond rapidly
C. The skin becomes permanently thicker
Answer: ☐A ☐B ☐C ☐D

14. A vaccine with high effectiveness can reduce disease spread because:
A. Fewer susceptible people and/or fewer infections can reduce opportunities for transmission
C. It removes every pathogen from the environment
Answer: ☐A ☐B ☐C ☐D

15. Which statement best integrates immunity and public health?
A. Individual immune protection and population-level transmission patterns can influence one another
C. Only antibodies matter for disease control
Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

B. The first vaccine permanently disappears from the body after hours
D. It removes all immune cells

Intermediate • 2 pts

B. Viruses are larger than bacteria
D. Viruses have cell walls that antibiotics cannot reach

Intermediate • 2 pts

B. All microbes become resistant simultaneously
D. Resistance cannot be inherited

Intermediate • 2 pts

B. Complete absence of immune cells
D. Only allergic responses

Intermediate • 2 pts

B. A bacterial disease
D. A genetic mutation in every immune cell

Intermediate • 2 pts

B. Red blood cells
D. Osteocytes

Advanced • 2 pts

B. Always indicates an autoimmune disease
D. Occurs only in the brain

Advanced • 2 pts

B. Pathogens always become weaker
D. Red blood cells remember antigens

Advanced • 2 pts

B. Vaccinated people cannot carry any microorganism
D. It guarantees lifelong sterilising immunity in every person

Advanced • 2 pts

B. Public health has no biological component
D. Vaccination affects only the vaccinated person's genes

Section B — True / False

20 marks

16. The immune system can distinguish many molecular patterns using specialised receptors.
A. True ☐ **B.** False ☐

17. Innate immunity is completely absent until after vaccination.
A. True ☐ **B.** False ☐

18. Antibodies are proteins that can bind specific targets.
A. True ☐ **B.** False ☐

19. All inflammation is harmful.
A. True ☐ **B.** False ☐

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

Basic • 2 pts

20. Vaccines are designed to induce immune memory against particular targets. Intermediate · 2 pts
A. True ☐ **B.** False ☐
21. Antibiotics are effective treatments for every viral infection. Intermediate · 2 pts
A. True ☐ **B.** False ☐
22. Antimicrobial resistance can spread through microbial populations. Intermediate · 2 pts
A. True ☐ **B.** False ☐
23. Autoimmune reactions can involve immune recognition of self molecules as targets. Intermediate · 2 pts
A. True ☐ **B.** False ☐
24. A person with immune memory can never become infected again. Intermediate · 2 pts
A. True ☐ **B.** False ☐
25. Herd or community protection can depend on how many people are susceptible and how efficiently an infection spreads. Intermediate · 2 pts
A. True ☐ **B.** False ☐

Section C — Fill in the Blank

20 marks

26. The first line of physical defence includes the skin and ____ membranes. Basic · 2 pts
 Answer: _____
27. The branch of immunity that includes rapid, broad defences is ____ immunity. Basic · 2 pts
 Answer: _____
28. Cells that secrete large amounts of antibodies are called ____ cells. Basic · 2 pts
 Answer: _____
29. Long-lived cells that contribute to faster secondary responses are ____ cells. Basic · 2 pts
 Answer: _____
30. A substance or molecular structure recognised by specific immune receptors is an _____. Intermediate · 2 pts
 Answer: _____
31. A vaccine aims to generate protective immune _____. Intermediate · 2 pts
 Answer: _____
32. The inappropriate immune response to harmless substances seen in hay fever is an _____. Intermediate · 2 pts
 Answer: _____
33. Immune attack against the body's own tissues is characteristic of ____ disease. Intermediate · 2 pts
 Answer: _____
34. Resistance of microbes to antimicrobial drugs is called antimicrobial _____. Intermediate · 2 pts
 Answer: _____
35. The movement of an infection through a population depends partly on transmission and the number of ____ people. Intermediate · 2 pts
 Answer: _____

Section D — Column Matching

10 marks

36-40. Match each term to its correct definition.

Column A	Answer	Column B
Antibody	_____	A. Supports a faster secondary immune response
Memory cell	_____	B. Exposure designed to train immune memory
Vaccine	_____	C. Immune attack against self
Autoimmunity	_____	D. Reduced susceptibility of a microbe to a drug
Antimicrobial resistance	_____	E. Specific protein that binds an antigen

Section E — Short Answer

20 marks

41. Compare innate and adaptive immunity in terms of speed, specificity and memory. Advanced · 4 pts
- _____
- _____
- _____
- _____

42. Explain why vaccination can protect an individual without giving them the disease the vaccine is designed to prevent. Advanced · 4 pts

43. Explain how antimicrobial resistance can increase in a bacterial population after antibiotic exposure. Advanced · 4 pts

44. Why does autoimmune disease differ conceptually from infection? Advanced · 4 pts

45. Explain how individual vaccination and population transmission can interact. Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Result / Grade: _____

Checked By: _____

Date Checked: _____

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Biology Comprehensive Practice Test 2 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. A | 9. A |
| 2. A | 10. A |
| 3. A | 11. A |
| 4. A | 12. A |
| 5. A | 13. A |
| 6. A | 14. A |
| 7. A | 15. A |
| 8. A | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. False | 21. True |
| 17. True | 22. False |
| 18. False | 23. False |
| 19. True | 24. True |
| 20. False | 25. False |

Section C — Fill in the Blank

- | | |
|-------------|-----------------|
| 26. mucous | 31. memory |
| 27. innate | 32. allergy |
| 28. plasma | 33. autoimmune |
| 29. memory | 34. resistance |
| 30. antigen | 35. susceptible |

Section D — Column Matching

- | | |
|---------------------|----------------------------------|
| 36. Antibody — E | 39. Autoimmunity — C |
| 37. Memory cell — A | 40. Antimicrobial resistance — D |
| 38. Vaccine — B | |

Section E — Short Answer (Model Answers)

- 41.** compare; rapid; broad; innate; defences
- 42.** controlled; antigen; exposure; immune; recognition
- 43.** existing; variation; acquired; resistance; selection
- 44.** autoimmunity; involves; immune; recognition; attack
- 45.** link; reduced; susceptibility; infection; individuals